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Executive Summary of the Thesis

The Impact of Windowing on EEG-Based Emotion Recognition: Pseudoreplication and Label Noise

Laurea Magistrale in Computer Science and Engineering - Ingegneria Informatica

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Academic year: 2025-2026

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1. Introduction

EEG-based emotion recognition promises real-time, objective measurement of affective state directly from neural signals. Progress in the field is largely driven by labeled datasets and by deep architectures capable of exploiting them, and a near-universal preprocessing step across these architectures is *windowing*: long experimental trials, each labeled once via post-hoc self-report, are cut into short, fixed-length windows that are then treated as independent training samples. This choice serves two practical needs: real-time applications require a prediction from a short signal segment rather than from an entire trial, and the number of trials collected per subject is typically small, so segmentation is used to multiply the number of available training samples. Windowing is inherited largely from general time-series processing, rather than derived from the temporal dynamics of affect itself, and it rests on two rarely tested assumptions.

Sample independence. Each window is treated as a statistically independent sample. This is violated if windows from the same trial share more variance with each other than windows of the same class from different trials, a pattern known as *pseudoreplication*.

Label stationarity. A single trial-level label is propagated to every window extracted from that trial, assuming the underlying affective state is constant for the entire trial duration: an assumption that is implausible a priori for experiences lasting tens of seconds, and a source of label noise whenever it does not hold. These assumptions are especially fragile in the affective domain: labels come from a single retrospective rating that is shaped by the experience's peak intensity and final moments (the peak-end rule); arousal (how intense the experience feels, from calm to excited) is intrinsically harder to introspect and report consistently than valence (how positive or negative it feels, from unpleasant to pleasant); and subjects exhibit stable, idiosyncratic response biases on the rating scale itself.

Research questions. Q1: Are windows from the same trial (and, more broadly, the same subject) geometrically almost indistinguishable, and does this depend on the distance metric used? **Q2:** How much label information is lost when moving from trial-level to window-level granularity, and does this translate into a measurable classification cost? **Q3:** Does a dataset whose trials are too short to window avoid these problems, and at what cost?

2. Related Work

The most directly related work is Gerster et al. [5], which analyses FACED, one of the datasets analyzed in this thesis, with a different methodology: supervised classifiers combined with stimulus-removal and label-substitution manipulations, rather than the geometric and correlational distance analysis used here. Their results show three of the six recommendations for neurophysiological state estimation proposed by Brouwer et al. [2] being violated in practice: classification is driven by confounds rather than by the affective state of interest, within-video temporal splits violate train/test independence, and stimulus decoding is not disentangled from emotion decoding. Khan et al. [8] show that this class of confound is not specific to a single dataset. Reviewing 101 highly-cited studies that use DEAP, they find that 87% exhibit at least one of four methodological pitfalls: segmentation leakage, biased feature selection, flawed hyperparameter tuning, or test-set-informed model selection. Under conventional k-fold cross-validation, reported accuracy climbs from chance to near-ceiling as the number of segments per trial increases; under leave-one-trial-out validation it stays flat instead, and the same climbing pattern reappears even on EEG-like data recorded from a phantom head with no neural activity. Brookshire et al. [1] report an even starker case of the same failure mode: EEG classification accuracy collapsing from 99.8% to chance once evaluation splits are changed from segment-based to subject-based. Together, these findings suggest that the concerns raised in this thesis are not specific to any single dataset, but likely affect EEG-based emotion decoding more broadly.

3. Methods

Let trial T_i carry a discrete label y_i, and let $w_{i,k}$, $k = 1, \ldots, K_i$, denote its windows ($K_i = 1$ for the unwindowed AFFEC dataset). Given a pairwise distance function $d_m(w_{i,k}, w_{j,l})$ under metric m, window pairs are grouped, restricting all comparisons to windows from the same subject, as *intra-trial* ($i = j$, $k \neq l$), *intra-label* ($y_i = y_j$, $i \neq j$), and *inter-label* ($y_i \neq y_j$), and per-subject group means $\bar{d}_{\text{intra-trial}}$, $\bar{d}_{\text{intra-label}}$, $\bar{d}_{\text{inter-label}}$ are aggregated across subjects. From these group means, two indices are defined: one probing pseudoreplication, the other class separability.

Trial Recognizability Index.

$$\text{TRI} = 1 - \frac{\bar{d}_{\text{intra-trial}}}{\bar{d}_{\text{intra-label}}}. \tag{1}$$

TRI $\to$ 1 indicates strong pseudoreplication (same-trial windows indistinguishable regardless of class); TRI = 0 indicates trial identity adds nothing beyond class membership.

Label Separability Index.

$$\text{LSI} = 1 - \frac{\bar{d}_{\text{intra-label}}}{\bar{d}_{\text{inter-label}}}. \tag{2}$$

LSI > 0 indicates genuine class separation; LSI is defined for every dataset, including AFFEC, whereas TRI is undefined where $K_i = 1\ \forall i$.

Both indices are computed under three distance metrics chosen to probe complementary representations. The Riemannian (AIRM) distance compares regularised channel-covariance matrices, capturing cross-channel connectivity structure. Dynamic Time Warping (DTW) compares the raw multichannel time series directly, aligning waveform shape while tolerating small timing shifts. The Wasserstein distance compares per-channel spectral power distributions, capturing how power is spread across frequencies independently of exact timing. Two non-geometric analyses complement the distance-based ones. First, the Pearson correlation between EEG band power and the self-reported rating is computed at trial level and at window level (with the trial's rating statically propagated to every window), for each of the five canonical frequency bands. Second, one-way ANOVA effect sizes (η^2 and its bias-corrected counterpart ω^2) are computed against the discretised class label at both granularities, providing a complementary measure of association that does not assume linear relationships between band power and rating. Finally, a classification pipeline (Differential Entropy features [4], i.e. the log-variance of the EEG signal in each frequency band, a standard descriptor of band power in EEG-based emotion recognition; linear SVM, regularisation strength selected by a nested, subject-blocked inner cross-validation) compares three conditions on the same cross-subject GroupKFold split: (1) trial-level, one feature vector and label per trial; (2) window-level, reported both as native per-window accu-

racy and, via majority vote across a trial's windows, as trial-level accuracy directly comparable to condition 1; (3) a one-window-per-trial ablation of condition 2, isolating whether any advantage of windowing comes from more independent trials or simply more training rows.

4. Experimental Design

Three datasets were used, summarised in Table 1. FACED [3] and DREAMER [7] are segmented into 1 s windows; AFFEC's [6] trials are short enough to be used unsegmented ($K_i = 1$ for every trial).

Table 1: Comparison of the three datasets analysed.

	FACED	DREAMER	AFFEC
Subjects	123	23	72
EEG channels	32	14	63
Sampling rate	250 Hz	128 Hz	256 Hz
Total trials	3,444	414	5,731
Windowable	yes	yes	no
Target	arousal	valence	arousal
Classes	3	2	3

The label used throughout is the subject's own post-hoc self-report, discretised into classes in a way that keeps the comparison meaningful across datasets that differ in scale and self-report format. For arousal (used for FACED and AFFEC), a three-class quantile split is used, so as to guarantee balanced classes regardless of the skew typically present in self-reported arousal; this yields near-perfectly balanced splits in both datasets. For DREAMER, valence is instead binarised with a fixed threshold at the scale midpoint, which produces a moderately imbalanced split. Classification uses subject-blocked K-fold cross-validation, with class-balanced weighting and no temporal smoothing, so that condition 2 is not artificially biased toward resembling condition 1.

5. Results

Subject and trial recognizability. Subject identity is strongly and consistently recoverable from raw EEG distance under every metric and every dataset tested independently of windowing. Trial identity is also recoverable, most strongly under the Riemannian metric (Table 2). Under the Riemannian metric, same-trial windows are on average 30–36% closer to one another than windows from different trials of the same class. The LSI, by contrast, sits at or within roughly two standard errors of zero for every one of the nine dataset×metric combinations tested (never exceeding 0.02 in absolute value), an order of magnitude smaller than TRI wherever TRI is defined. FACED-DTW is the one exception to the general pattern (TRI = 0.031); a channel-normalisation robustness check shows DTW loses 74–82% of its baseline TRI once amplitude differences are removed, against 26–57% for Riemannian and a markedly smaller effect for Wasserstein, consistent with DTW's baseline value being disproportionately driven by non-neural, session-specific amplitude artefacts rather than by a scale-invariant neural signature.

Table 2: TRI (mean ± SEM), baseline configuration. AFFEC is undefined by construction ($K_i = 1$).

Dataset	Riem.	DTW	Wass.
FACED	0.298 ± 0.008	0.031 ± 0.009	0.137 ± 0.005
DREAMER	0.361 ± 0.040	0.187 ± 0.049	0.120 ± 0.023

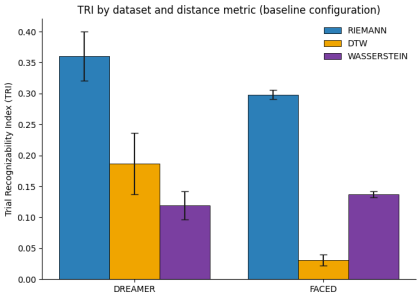


Figure 1: TRI for FACED and DREAMER datasets across metrics (mean ± SEM across subjects)

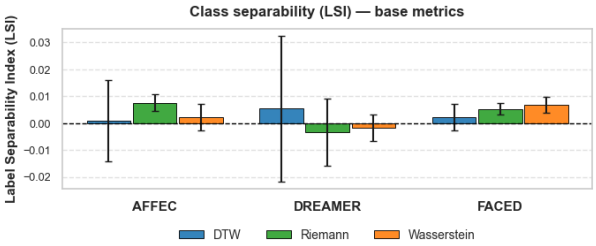


Figure 2: LSI (mean ± SEM across subjects) by dataset and metric.

Correlation collapse. The mean absolute

Pearson correlation between band power and the affective rating falls from trial level to window level in every single frequency band tested, by 50.5% overall in FACED and 46.3% in DREAMER (Table 3). The underlying trial-level correlations are themselves modest (no band exceeds $|r| = 0.06$), so the collapse should be read as windowing discarding roughly half of an already weak signal, not as destroying a strong one. AFFEC's trial-level correlation (0.017) sits in the same modest range as FACED's and DREAMER's *already-degraded* window-level values.

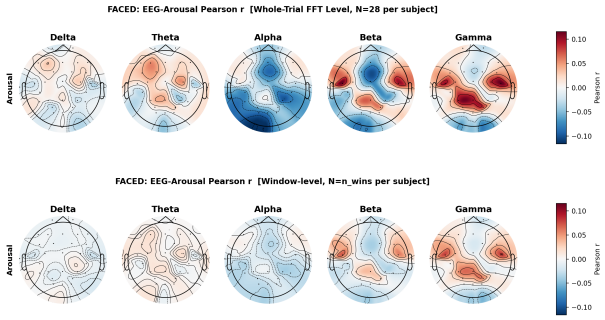


Figure 3: Pearson correlation for FACED, by frequency band: trial level (top) and window level, with the trial label statically propagated to every window (bottom).

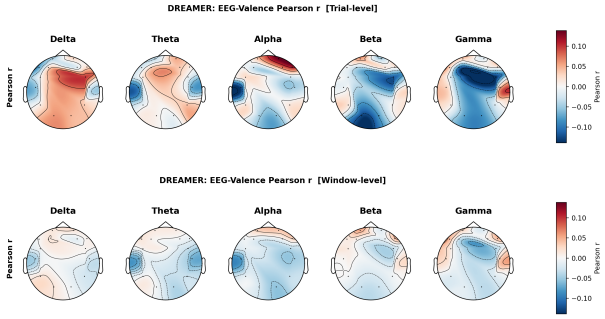


Figure 4: Pearson correlation for DREAMER, by frequency band: trial level (top) and window level, with the trial label statically propagated to every window (bottom).

Table 3: Mean absolute Pearson $|r|$ between band power and the affective rating.

Dataset	Trial	Window	% chg.
FACED	0.0302	0.0150	−50.5
DREAMER	0.0399	0.0215	−46.3
AFFEC	0.0173	–	–

The drop is even sharper for the omnibus ANOVA effect sizes: overall, raw η^2 falls by 81.5% (FACED) and 73.7% (DREAMER), while the more conservative, bias-corrected ω^2 falls by 62.1% and 45.1% respectively (Figure 5, Figure 6). Even at the trial level, before any windowing-induced degradation, ω^2 never exceeds 3.5% of variance in either dataset, and AFFEC's trial-level ω^2 (1.4%) sits in the same modest range.

Table 4: Mean η^2 (raw, upward-biased) and ω^2 (bias-corrected) effect sizes.

Dataset	Trial	Window	% chg.
η^2			
FACED	0.0780	0.0145	−81.5
DREAMER	0.0612	0.0161	−73.7
AFFEC	0.0333	–	–
ω^2			
FACED	0.0330	0.0125	−62.1
DREAMER	0.0279	0.0153	−45.1
AFFEC	0.0142	–	–

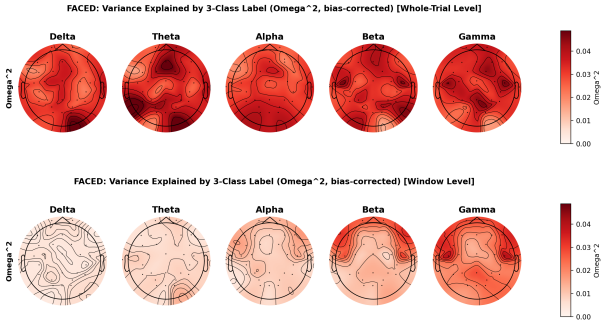


Figure 5: Bias-corrected ω^2 effect size of the label on EEG band power for FACED, by frequency band: trial level (top) and window level (bottom).

Classification. Once trial-level and window-level conditions are compared fairly, on the trial as the common final unit of prediction, windowing does not measurably cost accuracy in either dataset (Table 5): majority-vote window-level accuracy is, if anything, numerically higher than trial-level in both FACED (0.375 vs. 0.368) and DREAMER (0.579 vs. 0.558). The same picture holds under macro F1 (Table 5), the relevant metric for DREAMER given its moderate class imbalance: window-level F1 exceeds trial-level there too (0.552 vs. 0.537). The one-window-per-trial ablation never exceeds the full

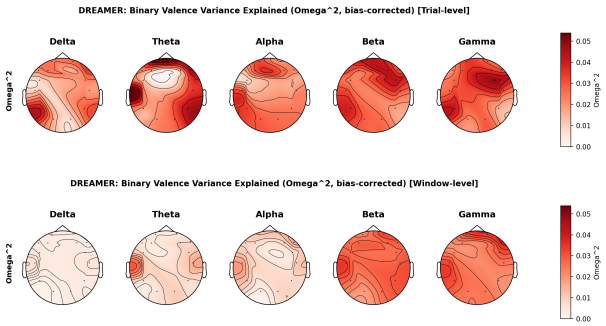


Figure 6: Bias-corrected ω^2 effect size of the label on EEG band power for DREAMER, by frequency band: trial level (top) and window level (bottom).

Table 5: Cross-subject K-fold classification accuracy and macro F1 (linear SVM on Differential Entropy features). Macro F1 is the relevant metric for DREAMER specifically, since its class split is moderately imbalanced.

Dataset	Trial-level	Window (maj.)	One window /trial
Accuracy			
FACED	0.368	0.375	0.364
DREAMER	0.558	0.579	0.547
AFFEC	0.359	–	–
Macro F1			
FACED	0.358	0.364	0.331
DREAMER	0.537	0.552	0.510
AFFEC	0.341	–	–

trial-level condition in either dataset (FACED: 0.364 accuracy; DREAMER: 0.510 F1), consistent with the number of independent trials, rather than raw training rows, bounding achievable performance. With balanced class weighting, no dataset's trial-level margin over chance exceeds its own fold-to-fold standard deviation (FACED accuracy: 0.035 vs. 0.042; DREAMER F1: 0.037 vs. 0.060; AFFEC accuracy: 0.026 vs. 0.034).

The AFFEC control. AFFEC eliminates windowing-induced pseudoreplication and label noise by construction, yet its trial-level classification accuracy remains close to chance, essentially indistinguishable from FACED's own trial-level accuracy on the same three-class arousal task. All three independent lines of evidence (geometric, correlational, and classification) agree that AFFEC shows little to no detectable class signal, pointing to a deeper, task-related difficulty in decoding arousal from these representations, independent of windowing.

6. Discussion and Implications

The trial-to-window correlation collapse is the best-evidenced finding: it is a standard statistic, it holds uniformly across every band and both windowable datasets, and it is not in tension with the classification result, since a ∼50% drop in a continuous correlation need not move a coarse, three- or two-way classification decision that is itself barely above chance. Two mechanisms could produce this drop: genuine label noise, from the trial's single rating no longer matching the affective state within each shorter window, or the higher estimation variance of a correlation computed on a shorter signal. Disentangling the two is difficult given the nature of these datasets: since only a single trial-level rating is collected, no window-level ground truth exists against which to test whether the affective state actually varies within a trial, so the two mechanisms cannot be separated without collecting finer-grained annotations. This thesis does not disentangle them, but the drop's consistency across every band and across two statistics with different assumptions (Pearson r and the ANOVA-based effect sizes) argues against either mechanism alone being negligible. These results carry a direct consequence for sample complexity. Because windows from the same trial are not independent, the trial count, not the far larger window count, is the more defensible estimate of the effective sample size (N_{eff}) of a windowed EEG dataset; FACED's 103,320 windows correspond to only 3,444 trials, and DREAMER's 24,840 windows to only 414, counts that are comparable to, or smaller than, the parameter counts of even compact architectures such as EEGNet. This does not invalidate any specific published result on its own, but it argues for reading reported sample sizes, and the model capacity they are implicitly assumed to support, with considerable caution. Rather than a blanket recommendation against windowing, which does not measurably cost accuracy once trial-level aggregation and subject-blocked evaluation are enforced, the thesis supports a narrower one: evaluation protocols should state explicitly the unit of label assignment, the granularity at which accuracy is measured, and the

use of subject-level splits; and N_{eff} should be counted at the trial level, not the window level, itself read as an upper bound given the further, subject-level confound.

7. Conclusions

Windows from the same trial are geometrically far more recognisable than the class label itself, most strongly under the Riemannian metric and consistently across all three metrics tested (Q1). Trial-level label information collapses substantially under windowing, yet this loss does not translate into a measurable classification cost once trial-level and window-level conditions are compared fairly (Q2). Removing the windowing confound entirely, as AFFEC does by construction, does not by itself buy better decoding, isolating a further representational limitation independent of windowing (Q3). Read together, pseudoreplication and label noise emerge as two symptoms of a single design error: treating the window, rather than the trial or the subject, as the unit of independent observation. This is precisely what makes the finding easy to miss in practice: windowing looks free, since it does not cost measured accuracy, while quietly inflating the sample size a dataset is assumed to have. The gap this thesis documents, between FACED's 103,320 windows and its 3,444 trials, or DREAMER's 24,840 windows and its 414 trials, is not a peculiarity of these two datasets: it echoes the segmentation-driven inflation reported by Khan et al.[8] and the subject-level leakage documented by Brookshire et al. [1], suggesting a shared methodological blind spot across EEG-based affective computing rather than an isolated concern. Closing this gap does not require abandoning windowing, which this thesis shows to be largely harmless to measured accuracy once evaluation is done correctly. It requires reporting practices that state explicitly the unit of label assignment, the granularity at which performance is measured, and the use of subject-level splits, and that count N_{eff} at the trial level rather than the window level. Until that becomes standard practice, published sample sizes in this literature should be read as upper bounds on the evidence they actually provide, not as the evidence itself.

References

[1] Geoffrey Brookshire, Jake Kasper, Nicholas Blauch, Yunan Wu, Ryan Glatt, David Merrill, Spencer Gerrol, Keith Yoder, Colin Quirk, and Ché Lucero. Data leakage in deep learning studies of translational EEG. *Frontiers in Neuroscience*, 18:1373515, 2024.

[2] Anne-Marie Brouwer, Thorsten O. Zander, Jan B. F. van Erp, J. Elmerik Korteling, and Adelbert W. Bronkhorst. Using neurophysiological signals that reflect cognitive or affective state: Six recommendations to avoid common pitfalls. *Frontiers in Neuroscience*, 9:136, 2015.

[3] Jingjing Chen, Xiaobin Wong, Chen Huang, Xin Hu, Xinke Shen, and Dan Zhang. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023.

[4] Ruo-Nan Duan, Jia-Yi Zhu, and Bao-Liang Lu. Differential entropy feature for EEG-based emotion classification. In *2013 6th International IEEE/EMBS Conference on Neural Engineering (NER)*, pages 81–84, 2013.

[5] Moritz Gerster, Elizaveta Sirotina, Anton Orlovskii, Alexandra Hertz, Juliette Champaud, Domenico Guarino, and Silvia Tulli. Stimulus identity rather than emotion drives EEG classification on the FACED dataset. *bioRxiv*, 2026. Preprint.

[6] Meisam Jamshidi Seikavandi, Laurits Dixen, Jostein Fimland, Sree Keerthi Desu, Antonia-Bianca Zserai, Ye Sul Lee, Maria Barrett, and Paolo Burelli. AFFEC multimodal dataset, 2026.

[7] Stamos Katsigiannis and Naeem Ramzan. DREAMER: A database for emotion recognition through EEG and ECG signals from wireless low-cost off-the-shelf devices. *IEEE Journal of Biomedical and Health Informatics*, 22(1):98–107, 2018.

[8] Nazmun N. Khan, Taylor Sweet, Chase A. Harvey, Calder Knapp, Dean J. Krusienski, and David E. Thompson. The role of review process failures in affective state estimation: An empirical investigation of DEAP dataset. 2025. Preprint.